Article

Recording-Grouped Bearing Fault Diagnosis under Directional Operating-Condition Shifts on the MCC5 Electric-Drive Benchmark

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Abstract

Windowed machinery benchmarks can overstate diagnostic transfer when correlated windows from one acquisition recording cross data partitions. We establish a recording-grouped benchmark on the MCC5-THU Motor bearing subset and examine transfer direction, grouped-partition variability, feature membership, and input semantics. The study includes 84 recordings, 15,036 overlapping windows, and 254 numeric features; neural protocols use frozen recording roles and training-only z-score statistics. XGBoost achieves 0.9964 macro-F1 on the predefined source-recording split and 0.9777 ± 0.0356 across ten grouped partitions. Random forest achieves 0.9790 for speed-circulation-to-torque-circulation transfer and 0.9891 for 20-to-40 Nm transfer. Leave-one-speed-out performance is strongly directional: macro-F1 remains near 0.99 when 1000 or 2000 rpm is withheld but falls to 0.6921 when extrapolating from 1000/2000 to 3000 rpm. Recording-level mean-probability aggregation preserves this high-speed boundary at 0.7020 macro-F1 over 28 test recordings, with a worst-class recall of 0.1111; majority voting yields 0.6444 macro-F1 and zero worst-class recall. A raw-file audit resolves all 84 catalogued recordings to distinct SHA-256 hashes, while a descriptive acquisition-date audit identifies residual session-confounding risk. The contribution is a direction-aware, recording-grouped MCC5 benchmark, not evidence of specimen-independent, session-independent, or universal speed generalization.

Keywords: bearing fault diagnosis; electric-drive condition monitoring; recording-grouped evaluation; directional operating-condition transfer; MCC5

1. Introduction

Electric drives are central to manufacturing, transportation, and energy conversion, while bearing defects remain an important source of vibration, reliability loss, and unplanned downtime [1–4]. Data-driven bearing diagnosis now spans engineered vibration descriptors, motor-current analysis, convolutional and attention models, transfer learning, and domain generalization [5–9]. The central deployment difficulty is not classification under randomly mixed laboratory windows, but diagnosis when speed, load, operating mode, acquisition session, or machine changes.

Localized bearing defects influence mechanical and electrical measurements differently. Vibration contains impacts, resonances, sidebands, and envelope modulation, whereas motor current reflects electromechanical coupling and torque-related modulation [10–13]. Order-domain processing can align selected rotational components when speed varies, but its validity depends on pulse quality, angle resampling, amplitude normalization,

Received:
Accepted:
Published:

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 https://doi.org/10.3390/machines1010000

and construction of a common order grid [14–17]. More recent domain-generalization studies learn invariant representations across machines and operating conditions, often with considerably broader validation than a single laboratory split [18,19].

Evaluation provenance is equally consequential. Overlapping windows from one recording share sensor placement, background noise, control trajectory, and acquisition artifacts. Allowing those windows to cross data roles can inflate performance and underestimate uncertainty [20–24]. The MCC5 source article recommends recording-level separation before segmentation and training-side validation for unknown-condition evaluation [25,26]. This study does not present recording grouping as a new algorithm. Instead, it operationalizes that principle as an auditable benchmark and asks four questions: (i) how sensitive are results to grouped partition choice; (ii) are operating-condition shifts symmetric in direction; (iii) what remains after exact feature membership replaces ambiguous substring groups; and (iv) which measured inputs are associated with apparent fusion gains under matched train-only standardization?

The contributions are threefold. First, a frozen recording-grouped MCC5 benchmark is reported with explicit transfer directions and zero recording overlap across neural training, validation, and test roles. Second, repeated grouped partitions, probability and vote aggregation at recording level, and class-stratified recording bootstrap are used to separate optimization variability from data uncertainty. Third, exact feature and input-schema audits are combined with raw-file identity and acquisition-session audits so that model results are interpreted within the independence that the public metadata can actually support. The resulting paper is a benchmark and evidence-audit study, not a claim of universal model superiority.

2. Related Work

2.1. Vibration, current, and order-aware diagnosis

Traditional bearing diagnosis combines time statistics, spectral energy, envelope analysis, and characteristic-frequency reasoning. Traditional descriptors remain widely used alongside compact and physics-informed learning models, particularly when recording-level sample sizes are limited [27–30]. Current-based diagnosis offers non-invasive electrical evidence but is sensitive to controller, supply, and loading conditions [10–13]. Multisensor methods can exploit complementary signals, although benefit depends on branch definitions, scaling, and availability rather than the mere presence of more inputs [31–36].

Order tracking expresses spectral content relative to shaft rotation and is attractive for variable-speed diagnosis. Nevertheless, nominal frequency division, tachometer-assisted angular resampling, and truly normalized common-order grids are not equivalent. This distinction matters here because the existing engineered order fields include raw band sums and key-phase diagnostics; they are evaluated as explicit feature blocks and are not assumed to be speed invariant.

2.2. Cross-mode learning and benchmark design

Domain adaptation uses information from a target domain, while domain generalization learns from source domains without target access. Existing bearing methods include adversarial alignment, meta-learning, causal factorization, boundary-aware objectives, and physically informed representations [37–45]. Their scientific claims depend strongly on split unit, direction, machine identity, and whether target data inform selection. The present work evaluates supervised baselines under fixed grouped protocols; it does not claim unsupervised adaptation or state-of-the-art DG performance.

3. Materials and Methods

3.1. Dataset and diagnostic task

MCC5-THU Motor was acquired on a 2.2 kW three-phase asynchronous motor connected to a torque sensor, gearbox, magnetic-powder brake, and synchronous acquisition system [25]. Eight channels contain key phase, torque, triaxial drive-end vibration, and three-phase current at 12.8 kHz for 90 s per recording. The bearing-only task pools light and high severities into healthy, inner-race, outer-race, and rolling-element labels. Pooling severity makes this a fault-location task and introduces within-class severity heterogeneity.

Raw-file provenance was audited against the official dataset copy without redistributing signal samples. For every catalogued recording, SHA-256, byte size, data shape, relative dataset path, and the acquisition timestamp encoded in the public filename were recorded. Repeated timestamps were examined by streaming value-level comparison. Acquisition dates were cross-tabulated with class and protocol role; Cramer's V, normalized mutual information, and an in-sample date-majority agreement were computed only as descriptive indicators of possible session confounding, not as diagnostic test performance.

Table 1. MCC5 bearing-only data composition.

Item	Value	Interpretation
Source recordings	84	24 rolling-element; 23 inner-race; 25 outer-race; 12 healthy
Overlapping windows	15,036	1.0 s window; 0.5 s stride; 179 windows per recording
Engineered feature table	260 columns	254 audited numeric candidates plus identifiers/labels
Signal channels	8	Key phase, torque, triaxial vibration, three-phase current
Sampling	12.8 kHz, 90 s	Synchronous acquisition for each source recording
Physical bearing-instance identity	Not available in public metadata	Recording separation does not establish specimen independence
Raw-file identity audit	84 distinct SHA-256 values	Excludes byte-identical files, not shared specimens or sessions
Acquisition dates	5	Date/class association is treated as descriptive confounding evidence

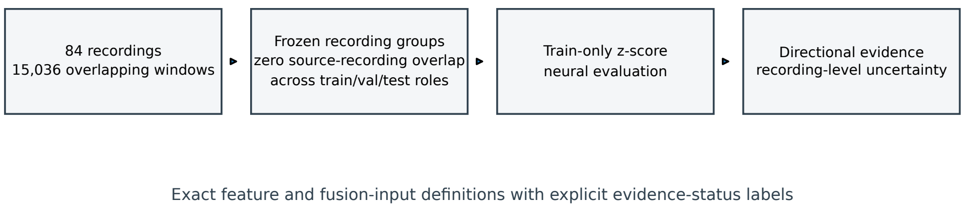

Figure 1. Overview of the recording-grouped MCC5 benchmark, from provenance-preserving segmentation and frozen recording roles to train-only preprocessing, directional evaluation, and recording-level uncertainty analysis.

3.2. Segmentation, feature schema, and raw inputs

Each 90 s recording produces 179 one-second windows with a 0.5 s stride. Adjacent windows overlap by 50%, so 15,036 windows are not treated as 15,036 independent experimental units. Engineered descriptors use all 12,800 samples. The audited feature table contains 254 numeric candidates: vibration time (36), vibration frequency (24), vibration envelope (15), vibration nominal order (27), vibration key-phase order/diagnostics (39), current time (36), current frequency (24), current nominal order (27), torque time (12), key-phase time (12), and RPM/load context (2).

The predefined raw-sequence protocol retains the first 8,192 samples (0.64 s); no resampling is performed. This input length was frozen before the standardized rerun to preserve architectural and computational comparability with the predefined neural

https://doi.org/10.3390/machines1010000

benchmark. A matched sensitivity retains all 12,800 samples and is reported as a duration analysis rather than a post-hoc replacement of the primary protocol. Sequence statistics are fitted per channel on training windows only. Scalar missing values are imputed by training medians and standardized per feature with training means and standard deviations. Validation and test sets use the stored training statistics unchanged. Near-constant scales are replaced by one. Automated audits require transformed nonconstant training means within 10^{-3} of zero and standard deviations within 10^{-2} of one.

3.3. Recording-grouped protocols

All neural roles are frozen at source-recording level. The existing source-recording validation set is preserved; the other training sides allocate approximately 20% of recordings per class to validation with seed 42. Windows from one source cannot cross roles, and no window-level fallback is permitted. The 3000-rpm test contains 28 recordings: 8 rolling-element, 7 inner-race, 9 outer-race, and 4 healthy.

Table 2. Frozen recording-grouped neural evaluation roles.

Protocol	Train	Validation	Test	Train/val/test overlap	Window fallback
Predefined grouped source-recording	59	13	12	0 / 0 / 0	No
Speed-circulation → torque-circulation	34	8	42	0 / 0 / 0	No
20 → 40 Nm	34	8	42	0 / 0 / 0	No
1000/2000 → 3000 rpm	45	11	28	0 / 0 / 0	No

Overlap entries are counts of shared source recordings across train/validation, train/test, and validation/test roles.

The predefined directions are speed-circulation to torque-circulation (cross-mode), 20 to 40 Nm (cross-load), and 1000/2000 to 3000 rpm. Robustness checks reverse mode and load directions and evaluate each leave-one-RPM-out direction. Ten class-stratified recording-grouped partitions were generated with split seeds 42–51 while the classical-model seed was fixed at 42. Within each diagnostic class, approximately 70%, 15%, and 15% of recordings were assigned to training, validation, and test roles; class-wise rounding yielded 59, 13, and 12 recordings, respectively, in every partition, with all four classes represented in each held-out test set. The fixed classical models used the training role and were evaluated once on the test role; the validation role was retained for protocol comparability.

3.4. Models and fusion-input definitions

Classical baselines include logistic regression, RBF-SVM, random forest, MLP, and XGBoost using training-side median imputation. Raw CNN, temporal convolutional network (TCN), and Transformer baselines use class-weighted cross-entropy, Adam with learning rate 10^{-3}, batch size 64, at most 30 epochs, and patience 8. Each neural comparison uses the same frozen internal validation map. All preprocessing statistics and model-selection decisions were fitted exclusively on the training and internal-validation recordings. Classical and neural hyperparameters were fixed before final test evaluation; the held-out test recordings were not used for scaling, early stopping, or hyperparameter selection. Full settings are reported in Supplementary Table S7 and the accompanying software supplement.

The branch network has vibration and current encoders, an optional scalar multilayer perceptron, sigmoid gating, and a classifier head. Six input definitions are distinguished: no scalar inputs; exactly RPM and load; a stored 26-field auxiliary bundle containing 24 waveform summaries (12 torque and 12 key-phase) plus two modality-presence indicators that are constant in this complete-modality subset; an auxiliary-context-28 bundle that adds RPM/load; a vibration-current network with a strict 254-feature engineered scalar branch (strict-254 scalar-branch fusion); and a legacy 262-field branch that additionally includes four window-position and four modality-availability fields. The 26/28-field settings were identified after inspecting the 3000-rpm holdout and are reported only as post-hoc exploratory evidence in the Supplementary Material.

3.5. Metrics and uncertainty

Macro-F1 is primary, worst-class recall identifies class collapse, and accuracy is supporting. Three optimization seeds are used generally and five for the focused 3000-rpm input audit. Seed standard deviation describes training variability only. For recording-level evaluation, window-level class probabilities are averaged within each test recording before assigning the recording label; majority vote is reported as a second aggregation rule. Class-stratified bootstrap resamples complete recordings 2,000 times; overlapping windows are never sampled as independent units.

4. Results

4.1. Raw-file identity and acquisition-session structure

All 84 catalogue entries resolve to official raw files, and all 84 SHA-256 digests are distinct. One timestamp is shared by three filenames. Pairwise comparison across 1,152,000 rows and nine columns confirms a common time vector but different signal values for every pair; the anomaly is therefore not an exact file or signal duplication. This audit does not determine physical specimen identity.

The recordings span five acquisition dates, and class composition is uneven across them. An in-sample date-majority mapping agrees with 61 of 84 labels (0.7262), with Cramer's V of 0.7714 and normalized mutual information of 0.5989. These are descriptive association measures, not out-of-sample diagnostic scores. They identify residual session-confounding risk that recording grouping cannot remove.

4.2. Directional classical benchmark

Classical feature models remain strong on source-recording, cross-mode, and cross-load protocols. Across ten grouped source-recording partitions, XGBoost reaches 0.9777 $\pm$ 0.0356 macro-F1, with a range of 0.9092–0.9992. Direction changes the conclusion most strongly for speed. Holding out 1000 or 2000 rpm remains near 0.99, whereas holding out 3000 rpm reduces random-forest macro-F1 to 0.6921 and worst-class recall to 0.1099.

Table 3. Directional robustness of the fixed 254-feature random-forest baseline.

Held-out direction	Model	Macro-F1	Worst recall	Accuracy
Speed-circulation $\rightarrow$ torque-circulation	Random forest	0.9790	0.9167	0.9761
Torque-circulation $\rightarrow$ speed-circulation	Random forest	0.9379	0.8822	0.9307
20 $\rightarrow$ 40 Nm	Random forest	0.9891	0.9795	0.9891
40 $\rightarrow$ 20 Nm	Random forest	0.9564	0.8457	0.9507
Hold out 1000 rpm	Random forest	0.9908	0.9860	0.9906
Hold out 2000 rpm	Random forest	0.9973	0.9951	0.9974
Hold out 3000 rpm	Random forest	0.6921	0.1099	0.6983

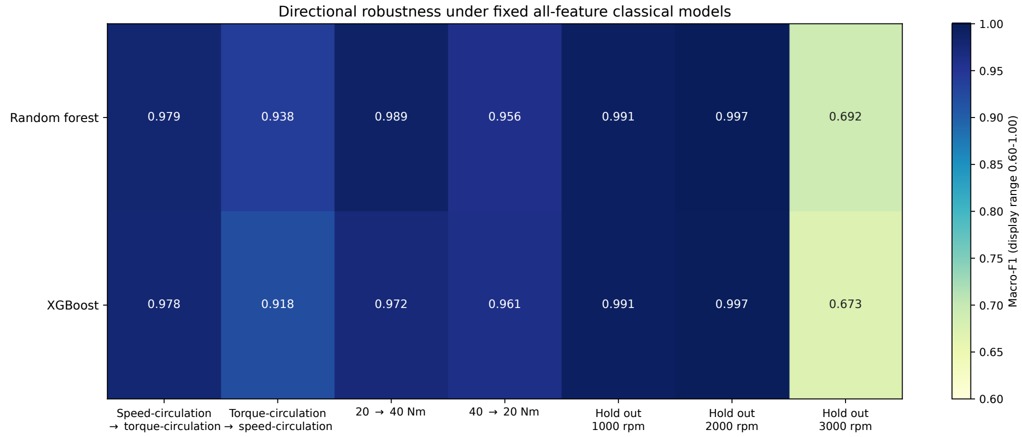

Figure 2. Directional robustness under fixed all-feature classical models. The displayed color range is restricted to 0.60–1.00 to make differences among the seven directional protocols visible.

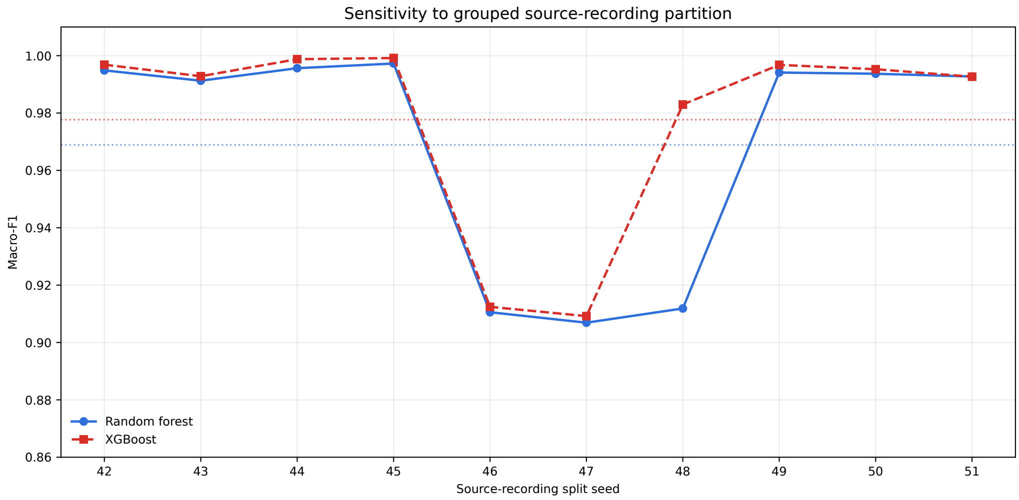

Figure 3. Sensitivity to ten class-stratified grouped source-recording partitions generated with split seeds 42–51. The classical-model seed is fixed at 42; variation therefore reflects partition choice rather than optimization randomness. Random forest uses a solid line with circular markers, whereas XGBoost uses a dashed line with square markers. Dotted horizontal lines denote the model-specific means across the ten partitions, and the vertical axis is restricted to 0.86–1.01. Each partition contains 59 training, 13 validation, and 12 test recordings, with all four classes represented in the test role.

4.3. Exact feature definitions

An exact membership audit shows that the former substring-defined groups mixed time, frequency, envelope, and order fields. Those legacy rows are retained only as provenance. Under exact grouping on the 3000-rpm holdout, XGBoost reaches 0.7569 with all non-order signal descriptors, 0.7556 after RPM/load is added, and 0.7596 after nominal-order fields are added. Full key-phase/order inclusion reduces the all-feature XGBoost score to 0.6733. Thus, non-order multisignal descriptors carry most aggregate high-speed performance, while the current order implementation does not establish speed invariance.

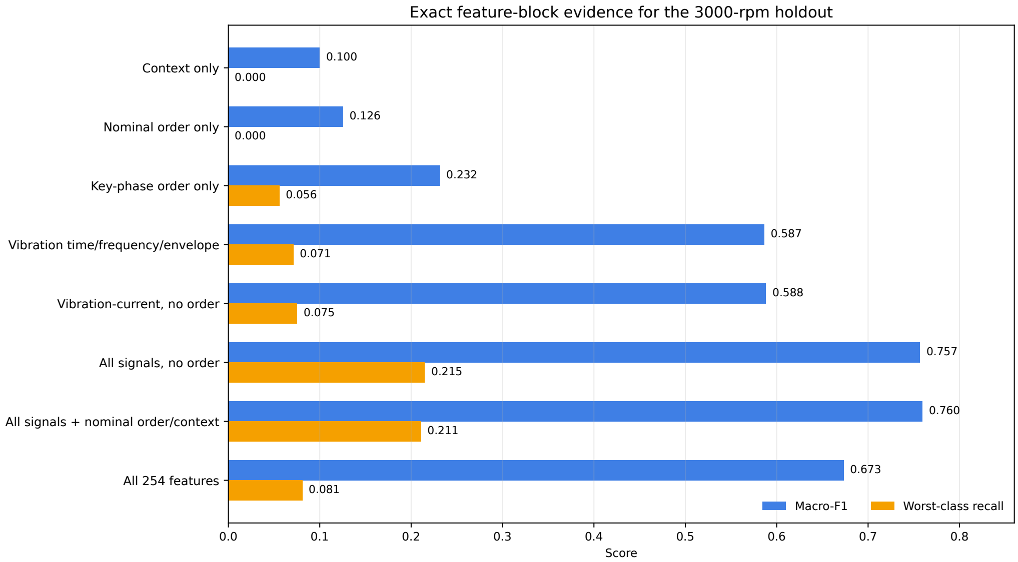

Figure 4. Exact feature-block evidence for the 3000-rpm holdout. Numeric labels report macro-F1 and worst-class recall for every feature definition, including zero-valued worst-class recall.

4.4. *Raw neural baselines*

Under train-only per-channel standardization, the Transformer reaches 0.9531 $\pm$ 0.0289 macro-F1 on source-recording evaluation. Cross-mode CNN, TCN, and Transformer means are all between 0.9356 and 0.9501. At 3000 rpm, however, CNN falls to 0.6644 $\pm$ 0.1523; TCN and Transformer fall further. Validation macro-F1 remains close to one, highlighting the mismatch between known-condition validation and unseen high-speed test data.

Table 4. Raw vibration-current models with train-only per-channel standardization.

Protocol	Model	Macro-F1 mean $\pm$ SD	Worst recall mean $\pm$ SD	Seeds
Source-recording	CNN	0.8535 $\pm$ 0.0286	0.5768 $\pm$ 0.0631	3
Source-recording	TCN	0.8759 $\pm$ 0.0263	0.5964 $\pm$ 0.0840	3
Source-recording	Transformer	0.9531 $\pm$ 0.0289	0.8762 $\pm$ 0.0804	3
Cross-mode	CNN	0.9501 $\pm$ 0.0076	0.8703 $\pm$ 0.0506	3
Cross-mode	TCN	0.9432 $\pm$ 0.0115	0.8749 $\pm$ 0.0367	3
Cross-mode	Transformer	0.9356 $\pm$ 0.0092	0.8915 $\pm$ 0.0320	3
Cross-load	CNN	0.8750 $\pm$ 0.0256	0.7337 $\pm$ 0.0839	3
Cross-load	TCN	0.7302 $\pm$ 0.0989	0.3726 $\pm$ 0.1638	3
Cross-load	Transformer	0.9076 $\pm$ 0.0602	0.7981 $\pm$ 0.1276	3
1000/2000 $\rightarrow$ 3000 rpm	CNN	0.6644 $\pm$ 0.1523	0.4270 $\pm$ 0.2296	3
1000/2000 $\rightarrow$ 3000 rpm	TCN	0.5011 $\pm$ 0.0465	0.2381 $\pm$ 0.1018	3
1000/2000 $\rightarrow$ 3000 rpm	Transformer	0.4703 $\pm$ 0.0810	0.1932 $\pm$ 0.0463	3

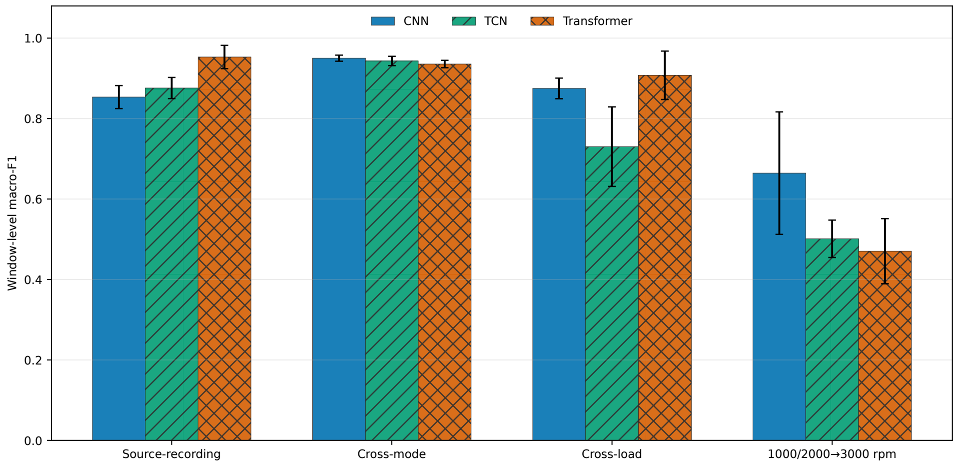

Figure 5. Raw-model macro-F1 with train-only standardization. Error bars are optimization-seed standard deviations ($n = 3$); bar hatching distinguishes model families in grayscale.

4.5. Fusion-input audit

With train-only standardization, strict-254 scalar-branch fusion reaches 0.9939, 0.9723, and 0.9299 mean macro-F1 on source-recording, cross-mode, and cross-load protocols, respectively. It falls to 0.4001 $\pm$ 0.0813 on the 3000-rpm holdout, where its worst-class recall is near zero.

The predefined focused input audit separates signal roles. No-scalar and RPM/load-only branches reach 0.7081 and 0.7124 mean macro-F1, respectively, so nominal RPM/load alone does not materially change the high-speed result under this matched protocol. The 26/28-field bundles were selected after viewing the same holdout; all of their numerical evidence is therefore confined to the Supplementary Material as hypothesis-generating disclosure.

Table 5. Predefined input-definition audit for the 1000/2000-to-3000-rpm holdout. Values are means and standard deviations across five optimization seeds. Post-hoc 26/28-field configurations are excluded and reported only in the Supplementary Material.

Configuration	Scalar inputs	Macro-F1 mean $\pm$ SD	Worst recall mean $\pm$ SD	Evidence status
Vibration-current; no scalars	0	0.7081 $\pm$ 0.0986	0.4324 $\pm$ 0.1507	Matched protocol
Vibration-current + RPM/load	2	0.7124 $\pm$ 0.0529	0.4272 $\pm$ 0.0964	Matched protocol
Strict-254 scalar-branch fusion	254	0.4001 $\pm$ 0.0813	0.0257 $\pm$ 0.0045	Matched protocol

4.6. Recording-level evidence and input duration

Recording aggregation exposes class-level fragility that window counts can obscure. Table 6 reports the primary classical model for every predefined protocol. Mean-probability aggregation remains perfect on the predefined source-recording and cross-load tests and reaches 0.9791 macro-F1 for cross-mode transfer. At 3000 rpm, however, it falls to 0.7020 macro-F1 with a worst-class recall of 0.1111; majority voting falls to 0.6444 macro-F1 and zero worst-class recall.

Table 6. Classical source-recording evaluation. Confidence intervals are class-stratified 95% recording-bootstrap intervals over 2,000 replicates. Majority vote is reported as a sensitivity aggregation.

Protocol	Model	Test recordings	Mean-probability macro-F1	95% interval	Majority-vote macro-F1	Mean-probability worst recall
Predefined source-recording	XGBoost	12	1.0000	[1.0000, 1.0000]	1.0000	1.0000
Cross-mode	Random forest	42	0.9791	[0.9365, 1.0000]	0.9791	0.9167
Cross-load	Random forest	42	1.0000	[1.0000, 1.0000]	1.0000	1.0000
1000/2000 to 3000 rpm	Random forest	28	0.7020	[0.6355, 0.8000]	0.6444	0.1111

For the corrected neural and fusion models, strict-254 scalar-branch fusion has zero recording-level worst-class recall on every seed at 3000 rpm. All values in Table 7 are based on 28 test recordings rather than thousands of correlated windows; post-hoc auxiliary-bundle results remain in the Supplementary Material.

Table 7. Recording-level mean-probability aggregation on the 3000-rpm holdout. Raw-CNN values use three optimization seeds; focused fusion-branch values use five. Means and standard deviations are calculated after aggregation within each recording; recording-cluster bootstrap intervals are reported separately in the Supplementary Material.

Model/configuration	Test recordings	Seeds	Recording macro-F1 mean ± SD	Recording worst recall mean ± SD
Raw CNN	28	3	0.7233 ± 0.2492	0.4722 ± 0.4111
No-scalar branch	28	5	0.7122 ± 0.1238	0.4000 ± 0.2054
RPM/load branch	28	5	0.7306 ± 0.0917	0.4000 ± 0.1369
Strict-254 scalar-branch fusion	28	5	0.3660 ± 0.1254	0.0000 ± 0.0000

Retaining the complete one-second raw sequence raises CNN window-level macro-F1 from 0.6644 to 0.7187 and reduces seed variability, but the high-speed boundary remains. This matched sensitivity isolates retained duration without changing architecture or split; it does not replace the frozen 8,192-sample primary protocol after observing the test result.

Table 8. Matched raw-CNN input-length sensitivity on the 3000-rpm holdout.

Retained samples	Macro-F1 mean ± SD	Worst recall mean ± SD	Batched inference time (ms/window)
8192	0.6644 ± 0.1523	0.4270 ± 0.2296	0.0432
12800	0.7187 ± 0.0578	0.4986 ± 0.1657	0.0661

Inference time was measured on an NVIDIA GeForce RTX 5080 over one complete test-set pass per optimization seed, using batch size 64 and no dedicated warm-up. The timer includes batched DataLoader iteration, host-to-device transfer, softmax, and output transfer to the host, but excludes disk I/O and preprocessing. Values are means across three seeded passes.

5. Discussion

5.1. What recording grouping establishes

Recording grouping removes direct sharing of overlapping windows across data roles. It does not establish independent physical bearing specimens because the public metadata do not provide a verifiable bearing-instance identifier. The protocol is therefore described specifically as recording-grouped, without making a broader leakage-safety claim. Results are limited to the MCC5 bearing-only subset and should not be interpreted as cross-machine validation.

Raw-file hashing further excludes byte-identical recordings, including within the repeated-timestamp group, but it does not resolve shared specimen or session identity. The acquisition-date audit shows substantial class/date association. Because date is not a model input, this does not establish timestamp-driven classification; it instead demonstrates that nuisance session structure may remain embedded in the signals after recording-level separation.

5.2. Why upward high-speed extrapolation is different

The asymmetry among the leave-one-speed-out tests is the central engineering result. Interpolation to 2000 rpm and downward extrapolation to 1000 rpm remain easy for the engineered classical models, but upward extrapolation to 3000 rpm produces class collapse for several otherwise strong models. This is consistent with a support-boundary problem: the test speed lies beyond the training range, and spectral distributions, amplitudes, controller behavior, and signal-to-noise relationships can all change together.

5.3. Order-aware is not automatically speed invariant

Frequency-to-order mapping is physically meaningful, yet raw order-band sums and imperfect key-phase estimates do not guarantee a common invariant representation. The exact block results support only a narrow conclusion: nominal-order features offer a small aggregate increment in one configuration, while order-only and full key-phase-order blocks are weak. A stronger order claim requires pulse-quality gating, robust interval filtering, fixed samples per revolution, common-grid interpolation, and density-normalized bands.

5.4. What the fusion-input audit supports

The predefined input audit supports two distinctions. First, strict-254 scalar-branch fusion performs strongly on three protocols with train-only z-score scaling but remains weak on the 3000-rpm holdout; this does not support a general claim that multisource fusion is defective. Second, adding RPM/load to the vibration-current branch produces little change under the high-speed protocol. A separate post-hoc audit of stored 26/28-field bundles is disclosed only in the Supplementary Material. Two stored presence indicators are constant here, leaving 24 varying torque/key-phase summaries; same-holdout discovery precludes a confirmatory interpretation regardless of their observed performance.

5.5. Implications for benchmark reporting

MCC5 studies should report the source recording as the split unit, provide transfer direction, distinguish train-seed variability from recording-level uncertainty, list exact input columns, and disclose whether candidate selection touched the test condition. Window-level metrics remain useful for local diagnostic behavior, but they should be accompanied by recording-level results whenever windows overlap.

6. Limitations

The study uses one controlled rig and one bearing subset. Physical specimen identity is not available, severity levels are pooled, and no external machine or independent bearing set is harmonized sufficiently for confirmatory validation. Five acquisition dates are unevenly associated with class (Cramer's V 0.7714), so session confounding cannot be excluded even though dates are not model inputs and source recordings do not cross roles. The 26/28-field auxiliary configurations are post-hoc. Neural seed counts are modest, and classical directional experiments use fixed hyperparameters rather than nested optimization. The strict full branch still uses a compact architecture; its 3000-rpm failure does not imply that all fusion strategies will fail.

The engineered order features do not yet use a fully normalized common order grid, and key-phase quality requires stronger gating. Raw data remain in sensor-voltage units in the processed table. Future work should confirm auxiliary-signal findings on a locked dataset, add a protocol-matched modern domain-generalization baseline, and test independent devices or specimens.

7. Conclusions

This study provides a recording-grouped MCC5 bearing benchmark with explicit operating-shift direction and train-only standardization. Classical models remain strong on source-recording, cross-mode, and cross-load protocols, but repeated partitions qualify the near-perfect single-split result. Upward extrapolation from 1000/2000 to 3000 rpm is the dominant boundary across classical and neural evidence.

The input audit also clarifies the fusion interpretation. Strict-254 scalar-branch fusion is strong on easier protocols but collapses at 3000 rpm, while RPM/load alone adds little to the vibration-current branch. Post-hoc auxiliary-bundle results are retained only as transparent Supplementary evidence and do not support model superiority. Raw-file and acquisition-date audits further show why recording separation should not be conflated with specimen- or session-independent validation. The contribution is therefore a reproducible directional benchmark and input-definition audit, not proof of specimen independence, universal order invariance, or confirmatory auxiliary-fusion superiority.

Supplementary Materials: The supporting information includes the complete fusion-input protocol table, per-seed focused results, recording-level evidence, severity and per-recording probability audits, recording-cluster bootstrap intervals, model architectures and fixed hyperparameters, and a versioned software supplement containing machine-readable evidence and reproduction materials.

Author Contributions: Conceptualization, Y.L.; methodology, Y.L.; software, Y.L.; validation, Y.L.; formal analysis, Y.L.; investigation, Y.L.; data curation, Y.L.; visualization, Y.L.; writing—original draft preparation, Y.L.; writing—review and editing, Y.L.; project administration, Y.L. The author has read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable. The study uses machinery signals and involves no human or animal participants.

Informed Consent Statement: Not applicable.

Data Availability Statement: The raw MCC5-THU Motor data are publicly available through Mendeley Data, Version 1, at https://doi.org/10.17632/6s3dggj9mw.1 and are not redistributed. Versioned code, frozen source-recording split maps, exact input schemas, result tables, reproduction scripts, and a compact raw-file audit containing relative paths, shapes, and SHA-256 digests for all 84 catalogued recordings are publicly available at https://github.com/YOMMMO/mcc5-bearing-paper-review/releases/tag/review-v3.1.3.

Acknowledgments: During the preparation of this manuscript, the author used OpenAI ChatGPT and OpenAI Codex (service versions accessed in July and August 2026) for code review, language editing, and document organization. The author reviewed and edited all outputs and takes full responsibility for the content of this publication. All scientific analyses were executed locally, and all numerical claims were checked against saved artifacts.

Conflicts of Interest: The author declares no conflict of interest.

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